

# Jason Li . Jarvis Consulting

I am a University of Waterloo graduate with a Bachelor of Mathematics in Computational Mathematics and a Minor in Computing. My passion lies in data analysis and software, which I further pursued during my co-op terms. I have hands-on experience working as a Data Scientist and as a Data Engineer, where I built AI applications and solved business-driven problems. I am proficient in Python, SQL, Linux, and data frameworks such as pandas and PySpark, and I am skilled in translating data solutions into actionable insights. I am currently seeking Data Scientist and Data Engineer roles where I can apply myself and solve complex real-world problems. My proven ability to produce results, from winning a hackathon to pioneering projects as a team lead, demonstrates my ability as a valuable asset in any team. I look forward to the possibility of discussing my background in data science and engineering with you.

## Skills

**Proficient:** Python, AWS, Linux, SQL, Agile/Scrum

**Competent:** Java, Git, Pandas, R, PySpark

**Familiar:** GCP, Tableau, C++, Cloudera Data Science Workbench, Waterfall methodology

## Jarvis Projects

Project source code: [https://github.com/jarviscanada/jarvis\\_data\\_eng\\_Li](https://github.com/jarviscanada/jarvis_data_eng_Li)

### Cluster Monitor [GitHub]:

- Automated the collection and storage of hardware specifications and real-time resource usage data from a 10-node Rocky Linux cluster by implementing Bash scripts and Linux command-line tools.
- Built a centralized PostgreSQL database within a Docker container to persistently store monitoring metrics.
- Enabled capacity planning and performance analysis by designing a relational schema that supports complex SQL queries for infrastructure optimization insights.

### Core Java Apps [GitHub]:

- Twitter App: N/A
- JDBC App: N/A
- Grep App: Developed a Java application that mimics the Linux grep command, allowing users to search and extract lines matching a specified regular expression from files within a directory structure.

### Python Data Analytics [GitHub]:

- Collected and analyzed LGS online retail transaction data using Python, Pandas, and NumPy in Jupyter Notebook.
- Cleaned and explored sales data from a SQL dataset to identify customer purchasing patterns and product trends.
- Provided insights to help the marketing team design targeted promotions and campaigns to improve customer engagement.

### Spark [GitHub]:

- Designed and implemented a scalable data engineering solution for London Gift Shop (LGS) by migrating legacy Python-based analytics workflows to Apache Spark on Databricks and Apache Zeppelin, enabling distributed processing of large-scale transactional data across cluster environments.
- Built an end-to-end data pipeline using PySpark, SQL, and Python libraries (pandas, NumPy, matplotlib, SQLAlchemy) to ingest, transform, and analyze customer transactions, uncovering insights into purchasing behavior, cancellations, and geographic trends.
- Enabled data-driven marketing strategies by developing analytics-ready datasets and customer segmentation models, supporting targeted campaigns such as personalized promotions, event-based offers, and retention strategies to improve revenue growth and customer engagement.

### Databricks [GitHub]:

- Built an end-to-end data engineering pipeline in Databricks using Medallion Architecture (Bronze/Silver/Gold), ingesting financial transaction datasets (CSV/JSON) from Azure SQL Database and Azure Storage via JDBC, ADF, and Lakeflow Connect for scalable ELT processing.

- Designed and implemented a Databricks DLT (Delta Live Tables) pipeline to ingest daily stock market data from the Alpha Vantage API, applying bronze-to-gold transformations to produce cleaned datasets for price and volume trend analysis.
- Orchestrated scheduled data workflows using Databricks Jobs to automate ingestion, pipeline execution, and downstream table refreshes, allowing for repeatable analytics-ready datasets and dashboarding from gold-layer insights.

## Highlighted Projects

**AWS Hackathon:** Developed an AI-powered application for the Canadian Intellectual Property Office (CIPO) that automates IP infringement detection and creates detailed asset descriptions, winning 1st place in the Fall 2024 AI Hackathon. Scraped data from the CIPO website and stored images and text descriptions in AWS S3, then utilized the Hugging Face API to classify images and generate short text descriptions for each. Implemented a simple, user-friendly interface through Jupyter Notebooks and Python.

**AWS Datahub:** Led the creation of an AI-powered data hub using AWS S3, Bedrock, and Lambda to support non-technical teams. Supervised and distributed tasks while developing the backend with AWS Lambda, creating guardrails using AWS Bedrock, and designing the landing page. Scraped data from Outlook and stored tutorial videos and guides in AWS S3, delivering project goals ahead of schedule with excellent feedback that improved efficiency and client satisfaction.

**OKR Data:** Created Objective and Key Results (OKR) data for past and current fiscal years using Python, SQL, and Jupyter Notebook. Retrieved and processed large datasets from Apache Impala using SQL, performing in-depth data cleaning and analysis with Pandas and NumPy to generate actionable insights. Presented analytical findings to shareholders through Power BI visualizations and PowerPoint reports, and automated reporting processes with Python scripts, reducing manual effort and report turnaround time by 30%.

**VIP Heatmap:** Constructed a full-scale heat map of Ontario to visualize the geographic distribution of VIP clients. Cleaned VIP client data with SQL and Pandas by standardizing postal codes to the first 3 digits for analysis. Leveraged Excel's heat map visualization to map VIP client locations, providing stakeholders with clear insights into key target regions for strategic outreach and delivering actionable insights that guided decision-making.

## Professional Experiences

**Data Scientist, Jarvis (Jan 2026-present):** Built and automated a Linux cluster monitoring system that collects hardware specifications and real-time resource usage from multiple hosts, storing them in a centralized PostgreSQL database via Bash, Docker, and cron. Designed a relational PostgreSQL data model and developed analytical SQL queries to explore customer activity patterns, resource utilization, and cost drivers using business-oriented metrics. Analyzed multi-year e-commerce transaction data using Python (Pandas, NumPy) in Jupyter to uncover customer purchasing behaviour and support targeted marketing campaign strategies. Cut exploratory analysis time by 40% via automated data pre-processing and feature engineering.

**Data Scientist, Innovation, Science and Economic Development Canada (Sept 2024 – Dec 2024):** Developed an AI-powered application for the Canadian Intellectual Property Office that automates IP infringement detection and creates detailed asset descriptions, winning 1st place in the Fall 2024 AI Hackathon. Led the creation of an AI-powered data hub using AWS S3, Bedrock, and Lambda to support non-technical teams. Designed and delivered a comprehensive 6-week Python programming and data analysis course for non-technical professionals, covering fundamentals through advanced data manipulation with DataFrames. Constructed a data pipeline for weekly tasks using AWS Data Pipeline, saving 3+ hours of runtime weekly.

**Data Engineer, Ontario Lottery and Gaming Corporation (Sept 2023 – Apr 2024):** Created Objective and Key Results (OKR) data for past and current fiscal years using Python, SQL, and Jupyter Notebook. Improved automated jobs in Cloudera Data Science Workbench, resulting in a 25% speedup in runtime. Aggregated data using SQL and Python and presented insights using Excel, PowerPoint, and Power BI. Engineered a new mapping method to track valuable players by region using FSA data to facilitate special events, and provided data and touch-points to other teams as the Lead Analyst for multiple labs.

**Software Test Engineer, Infrastructures For Information (Sept 2022 – Dec 2022):** Designed and executed 300+ test cases using QaTraq, resulting in a 25% speedup in Portal application testing. Automated 500+ test cases using i4i's UAT automation features, resulting in a 20% reduction in resource usage. Pioneered a project to rewrite regression tests to improve efficiency, resulting in 10% fewer regression test cases. Identified, analyzed, and documented bugs in the Portal product and A4L application using QaTraq.

**Software Engineer, Natural Resources Canada (Jan 2022 – Apr 2022):** Standardized digitization of 1,000+ physical storm data records using RStudio and supervised machine learning. Improved storm recognition ML model prediction accuracy by 40% through cleaner training data preparation. Migrated a legacy codebase for a magnetic data application from Pascal to C++ to improve maintainability. Oversaw the creation of training documentation and became the subject matter expert on image processing software.

**Data Scientist, Sprint Data Systems (May 2021 – Aug 2021):** Gathered and analyzed 100,000+ usage and behavior data points by data mining client IP addresses and locations. Optimized data mining and cleaning scripts using Bash, resulting in a 30% speedup in processing time. Collected and standardized 30,000+ courier package data entries into specific formats using Excel. Created a data pipeline using Python to clean and identify inconsistent customer data for analytics.

## Education

**University of Waterloo (2020-2025),** Bachelor of Mathematics in Computational Mathematics, Minor in Computing, Mathematics - President's Scholarship

## Miscellaneous

- Retired Professional Swimmer/Varsity Swimming
- Gaming
- Reading